

Gnomon: Architecture and Operational Evidence for an IELTS Writing Feedback Engine*

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Abstract

This report documents Gnomon, a feedback engine for IELTS Writing architected so the model never outputs a grade: across five criteria, Task Achievement, Task Response, Coherence and Cohesion, Lexical Resource, and Grammatical Range and Accuracy, it extracts and quotes facts only, and a deterministic scorer maps those facts to a result. This is a systems and evidence report, not an empirical accuracy evaluation: it documents architecture, measured operational evidence, and infrastructure economics, and it makes no accuracy, agreement, or band-prediction performance claim for the engine described here.

The pipeline behind that engine is large and independently auditable. The training corpus holds 101,175 individually sourced essays as of this writing (99,487 at the last written snapshot; the live tool, not the prose, is the source of truth) drawn from 47 distinct sources. Real IELTS examiners' own marginal comments, 2,014 of them hand-written on real marked scripts, were mined, scrubbed of every raw band number, and distilled into 264 label-free per-criterion training facts. A separate weak-supervision pool of 223,600 essay-criterion label pairs, built from 57,078 filtered essays, was constructed and then deliberately never used for training, because every row carries a raw band label the architecture's label-free doctrine (LD1) forbids the model from learning. The production adapter fine-tunes a 27-billion-parameter open-weight vision-language model (Qwen3.8-27B) by LoRA SFT on a curated bundle of 9,825 rows, a checkpoint that exists on disk today.

Four evidence assets anchor the engine's own measured record. A human-rater ceiling study of 168 double-graded essays and 669 criterion pairs across four of the five criteria (Task Response excluded), mean MAE 0.36. A fail-closed measurement harness that correctly withheld a paid run over 2,232 selected criterion units, 629 essays, by returning HOLD when 366 payloads failed a vision or instruction check. A frozen 1,219-case policy classifier deck holding 51 false positives and zero false negatives against a 174/28 baseline. And QLoRA fine-tuning of that 27-billion-parameter model at a measured \$0.12 USD per million real tokens on the cheapest of three rental GPU rails tested. The current four-criterion engine MAE, matching the scope of the measured human-rater floor, is MISSING from the record and is not claimed here.

Separately from the current architecture's own unmeasured engine MAE, the project's development record holds measured per-criterion results from two earlier, now-superseded scoring mechanisms, reported here with full scope and caveats, not folded into the claim above. An earlier theta-nearest-neighbor scorer (session 132-134, June 2026) measured, on a 30-essay gated holdout of essays with precomputed ability coordinates, GRA MAE 0.083, LR MAE 0.100, CC MAE 0.083; its Task Achievement figure is itself unreconciled between two dated internal sources, 0.233 and 0.333, and is reported as a conflict, not a single number. Essays without a

*This is a systems-and-evidence report, not a peer-reviewed accuracy claim, and makes no accuracy claim about the scoring engine's IELTS band predictions.

precomputed coordinate fell back to a materially worse 0.6-0.8 MAE, so these figures describe roughly half the corpus’s easiest-case essays. A second, contemporaneous scorer for Task 2 measured, on a 157-essay production gate with source-specific calibration, TR MAE 0.319, CC MAE 0.268, LR MAE 0.248, GRA MAE 0.226 (all rho above 0.91); a targeted per-band calibration fix on a further 65-essay holdout cut TR MAE from a 0.446 baseline to 0.308. Both systems are architecturally distinct from, and predate, the label-free deterministic-Rasch pipeline this report documents; neither result is a measurement of the engine described above.

1 Introduction and positioning against AES practice

IELTS Writing is marked against public band descriptors: nine ordinal levels (0 to 9) defined separately for the two tasks and their constituent criteria [1]. Any automated system that touches this construct inherits an obligation to reproduce, or at least respect, that ordinal structure rather than collapsing it into a single learned number.

The dominant lineage in automated essay scoring (AES) does the opposite: it optimises directly against a score. Feature-based systems such as e-rater extract linguistic features, grammar, usage, mechanics, style, organisation, development, and feed them into a regression trained to reproduce a human-assigned score [2, 3]. End-to-end neural systems replace hand-engineered features with learned representations (LSTM, CNN, or embedding-based encoders) but keep the same objective: a model that outputs a score and is trained to minimise error against one [4, 5]. Both lineages are surveyed field-wide by Ke and Ng [6], and both are benchmarked on the same corpus, the Hewlett Foundation’s ASAP-AES set of eight essay prompts released as a Kaggle competition [7], scored by quadratic weighted kappa (QWK) [8, 9].

Gnomon departs from both lineages at the architecture level, not the tuning level. The generative model never regresses a score and never classifies a band. It is architected so that it extracts and quotes facts and reasoning about an essay only, and never answers MET/UNMET, picks a level, or emits a band or a score. Those extracted facts are consumed by a separate, deterministic program, a Rasch (item-response-theory) scorer that converts binary MET/UNMET item answers into a band, with item difficulty estimated label-free from the response matrix and floor items held non-compensatory. No grade label reaches the generative model’s output at any point in the pipeline: the model produces evidence, the scorer produces the result.

This split changes what “performance” means for the system. The metric arc adopted here is per-criterion mean absolute error (MAE), measured separately across all five criteria the engine spans, Task Achievement, Task Response, Coherence and Cohesion, Lexical Resource, and Grammatical Range and Accuracy, and this is the sole optimisation target, not QWK on a single blended score. Holistic correlation between the engine’s output and a reference score is computed and reported as a byproduct, but it is never treated as a gate for any experiment or design decision, because a single correlation coefficient can look strong while masking systematic per-criterion error, exactly the failure mode a construct-faithful, criterion-separated instrument is meant to avoid.

The choice of MAE-per-criterion over QWK-on-a-blend is grounded in the field’s own governing evaluation framework, which specifies that an automated scoring engine must demonstrate construct fidelity, reliability, agreement across multiple statistics, generalisability, and consequential validity, not correlation with a reference score alone [10].

This report is a systems and evidence report, not an empirical accuracy evaluation of the engine: no engine-level accuracy, MAE, agreement-with-examiner, or band-prediction figure is reported or claimed anywhere below. What is reported is what the record measures directly: the human-rater ceiling study, the harness validation counts, the classifier deck scores, training configuration and throughput, and, as operational context for how those measurements are produced and checked,

the fact-emission and judgement mechanisms the pipeline runs on (Sections 5 and 7 in particular describe mechanisms and controls, not engine-accuracy results). The remainder of this report follows the field’s own evaluation framework: it separates what the architecture guarantees by construction (Sections 2 to 4) from what has actually been measured (Sections 5 to 7), and states plainly, in Section 9, what is not yet measured and why that gap is reported rather than papered over.

2 Design: five criteria, task grounding, vision law, grade-label quarantine, deterministic scorer, abstain

Gnomon grades against the same five criteria and the same construct language as the IELTS Writing Band Descriptors and their plain-language companion: Task Achievement (TA), Task Response (TR), Coherence and Cohesion (CC), Lexical Resource (LR), and Grammatical Range and Accuracy (GRA) [1, 11]. The design decisions below (criterion coverage, task grounding, the vision requirement, the quarantine that keeps a grade out of every intermediate artefact, the deterministic scorer, and the abstain rule) all follow from the same starting commitment: reproduce the examiner’s construct, not a proxy for it.

2.1 Five criteria, confirmed in three independent pipeline stages

The optimisation target is per-criterion mean absolute error (MAE) summed across all five criteria. That is not only a stated target; it is load-bearing in the pipeline that builds training data. The supervised fine-tuning (SFT) head-assembly step maps trait columns for TR, CC, LR, GRA, and TA when constructing label-free training input, placing TR inside the same five-criterion set as the other four rather than as an addition made later. A second, independently written stage that converts mined human-examiner comments into training-eligible text covers the identical five criteria. A third stage, the per-trait band-anchor selector, enumerates six trait values: TR, CC, LR, GRA, TA, and GT_TA, where GT_TA is the Task 1 General Training letter-writing variant of Task Achievement, stored in the same column as TA. Three separately authored components agreeing on the same five-plus-one criterion set is the confirmation this report relies on, rather than a single declared list.

2.2 Task grounding: TA on the chart, TR on the prompt

The descriptors define TA as a Task 1 criterion and TR as a Task 2 criterion [1]. Gnomon enforces that split mechanically rather than by convention. TA carries a Task-1-specific, chart-construct vocabulary: the overlay keywords used to detect a chart-grounded response include overview, trend, feature, data, chart, graph, and figure. TR is Task-2-only and prompt-grounded, with no chart involved; one training-data source corpus of Task 2 model essays is used for four criteria but explicitly excludes TA, because that corpus has no chart and TA cannot be scored without one. Three traits, TA, TR, and GT_TA, require their source essay to be verified against the correct task before use, specifically to prevent a Task 1 essay contaminating a Task 2 label or the reverse.

2.3 Vision law: no image, no grade

Every chart-grounded item must carry the actual chart. At training-data admission, a TA row drawn from a Task 1 essay is blocked outright, marked BLIND, if no locally resolvable image passes a validity gate. That validity gate is singular: every code path that resolves an image for

training or inference must agree with the same file-size floor and structural image check, rather than each caller running its own opinion of what counts as a valid image. At measurement time the same law holds for the harness: every TA call attaches the real chart as an actual multimodal (base64) content block, and a call with no image attached is defined as not a valid TA grading call at all.

2.4 Grade-label quarantine

The generative model is prompt-forbidden from producing a grade in any form, MET or UNMET, a level, a band, or a score, including inside a free-text reason field. That prohibition is checked twice more, independently of the prompt. At training-data admission, a regex blocks any training completion where the word band or score is followed by a number. At the artefact level, a two-tier scanner checks every generation-lane file for a leaked label, one tier on value-carrying field names, one tier on a value-carrying regex pattern, and fails closed (a non-zero exit) on any hit. That scanner, and its harness counterpart, are built to match only value-carrying patterns, a band followed by a number, MET or UNMET followed by a value, a score followed by a number, never a bare prohibition word, so a prompt that names the word “band” in order to forbid it does not trip its own scanner. This firewall is a syntactic, regex-level check: it has not been tested against a paraphrased leak that avoids the literal tokens it matches on (for example, a description of an essay’s standing in words rather than symbols), and Section 9 discloses that gap rather than implying the firewall covers every leakage form.

2.5 Deterministic scorer

Facts extracted from the essay are converted to a band by a Rasch (item response theory) model, not by the generative model itself. Binary evidence items (met or not met) feed a latent-ability estimate; item difficulty is estimated from the response matrix itself, without reference to any grade label; and a small number of floor items are non-compensatory, meaning a failed floor item caps the resulting band regardless of how high the latent-ability estimate runs elsewhere.

2.6 Abstain: disagreement is the signal

The scorer produces more than one independent band estimate per criterion. When those estimates agree, the engine commits their median as the result. When they do not, the engine abstains and returns a typed reason rather than guessing.

Table 1: Abstain decision thresholds

Condition	Threshold	Action
Spread of estimates	at least 1.0 full band	abstain
Standard deviation of estimates	at least 0.5	abstain
Neither threshold crossed	not applicable	commit median

Disagreement among the independent estimates is treated as the uncertainty signal itself, not as noise to be averaged away. These two thresholds are configured design values; no abstain rate or coverage-versus-accuracy curve has yet been measured against them at production scale, and Section 9 states that gap rather than implying the operating point has been empirically tuned.

This fact-emit-then-score split follows the same shape as structured form-filling evaluation, where a model produces intermediate, checkable judgments rather than a single end score directly

[12, 13]. It is also the design answer to the governing requirement that an automated scoring system’s evidence be evaluated for validity and construct fidelity, not for correlation with a reference score alone [10].

3 Human-rater ceiling study (method, numbers, why it gates any accuracy claim)

3.1 Method

168 essays were double-graded by independent human examiners, producing 669 scorable criterion pairs across four of the five criteria: Task Achievement (TA), Coherence and Cohesion (CC), Lexical Resource (LR), and Grammatical Range and Accuracy (GRA). Task Response, the fifth criterion, is not part of this study’s scope, and no human-rater floor for TR exists in the record. For each criterion the two examiners’ band scores on the same essay were compared, and the mean absolute error (MAE) between the pair was taken as that criterion’s floor: the smallest disagreement two qualified examiners produce grading the same work, and therefore the lower bound any automated score can be judged against.

3.2 Results

Table 2: Human-rater ceiling study results

Criterion	n (pairs)	Floor MAE	Inter-rater r
TA	168	0.426	0.779
CC	168	0.393	0.778
LR	168	0.345	0.777
GRA	165	0.270	0.891
Mean (four criteria)	—	0.36	—

The same per-criterion figures and the mean floor of 0.36 MAE are independently restated in the product record, which describes this measurement as the strongest evidence asset held. A third record, the decision log, carries the same study again: 168 dual-graded essays, 669 criterion pairs, zero parse failures, per-criterion MAE matching the table above, inter-rater r ranging 0.78 to 0.89, and a Spearman-Brown two-rater-mean reliability of approximately 0.88.

3.3 Why this gates any accuracy claim

Two qualified human examiners grading the same essay do not agree exactly; the disagreement measured above, mean 0.36 MAE with r between 0.78 and 0.89, is what that disagreement costs on this corpus, across the four criteria measured. Consider a hypothetical engine scoring 0.36 MAE against one examiner on those four criteria: it would be agreeing with that examiner about as well as a second examiner would; there is no perfect human reference to compare it to, only two graders whose own agreement sets the ceiling any third scorer, human or automated, can reach. No such measurement exists for this engine at the time of writing, and none is claimed here.

This is the substantive reason this report makes no engine MAE claim. The governing framework for evaluating automated scoring of constructed-response tasks requires agreement to be assessed with more than one correlation statistic and against a documented reliability baseline before an

accuracy claim counts as evidence [10], a requirement grounded in the formal testing-standard validity and reliability evidence base [14]. For this study’s ordinal, multi-rater design, the statistics that framework calls for are quadratic-weighted kappa [15, 16] and an intraclass correlation computed under an explicitly named ICC form [17]; the MAE and Pearson r reported here are a first measurement, not a completed evaluation against that standard. This report deliberately does not adopt the field-standard convention of quadratic-weighted kappa scored on the ASAP-AES benchmark set [7] as its primary metric, since that convention was built for a different corpus and scale; the per-criterion floor measured here is the baseline this project’s own accuracy claims must be reported against instead.

Until an engine measurement exists, benchmarked against this floor under the statistics above, no engine-versus-human MAE claim is meaningful. None is made in this report: the current four-criterion production-engine MAE, matching this study’s own scope (TA, CC, LR, GRA), is MISSING in the record. No comparable human-rater floor exists for TR, and no five-criterion combined floor is claimed; any future comparison against the 0.36 figure is scoped to the same four criteria this study measured, not to the engine’s full five-criterion span, until a TR study exists.

4 Fail-closed evaluation harness (refusal codes, seeded negatives, the 2,232-unit run, HOLD)

No paid measurement run starts until a dedicated harness has checked every candidate payload against the two structural laws the rest of this report rests on: the vision law and the grade-label quarantine. A payload that fails either check is refused with a typed code, not silently dropped, and a batch containing refusals returns a HOLD decision rather than proceeding to spend. HOLD is reported below as evidence that this control mechanism worked as designed: it establishes that the harness correctly withheld a paid measurement run, not that any accuracy figure was produced.

4.1 What the harness checks at the payload level

Two checks run before any request leaves the machine:

- **Vision law.** A chart-grounded (Task Achievement) unit is valid only if its payload carries a real multimodal image content block; a text caption describing the chart is not accepted as a substitute.
- **Label quarantine.** The harness reuses, verbatim, the three value-carrying grade-label regular expressions already defined for the project’s label firewall: `band[:]=[0-9]`, `(UN)?MET[:]=\S+`, `score[:]=[0-9]`, rather than maintaining a second pattern set that could drift from the first.

4.2 Offline test coverage

The harness ships eight offline test functions, run at zero cost against fixtures rather than against a live API:

The plan document that specifies this harness records an earlier reproducibility check of “6 passed in 0.18s”; the test file as it stands today defines eight, two more than that recorded tally. That gap is flagged, not smoothed over: a test count is itself a claim, and an unreconciled six-versus-eight discrepancy is the kind of drift the harness exists to catch elsewhere.

Table 3: Harness offline test coverage

No.	Test	Proves
1	Image-slot requirement	A visual-input unit without an image block is rejected
2	Task-2 instruction requirement	Every Task-2 criterion unit without prompt text is rejected, not only Task Response units
3	Task-2 sidecar scope	The sidecar instruction source applies only where it applies
4	Visual-exclusion binding	Non-chart criteria are not wrongly forced through the vision check
5	Fixture MAE vs hand-computed truth	The MAE arithmetic matches a value worked out by hand
6	Denominator naming	Every reported rate states what it is a proportion of
7	Anomalous-floor halt	A score at the floor with high sub-scores stops the batch
8	Seeded negative case	A deliberately planted bad row is caught, proving the firewall can fail

4.3 The registered-frame dry run

Run against the real evaluation frame (2026-09-02), the harness selected 629 essays and 2,232 criterion units:

Table 4: Criterion units selected by the registered-frame dry run

Criterion	Units
TA	317
TR	214
CC	567
LR	567
GRA	567
Total	2,232

Payload validity split 1,866 valid against 366 invalid, with every invalid unit carrying a typed refusal code:

`MISSING_TASK2_INSTRUCTION` exceeds the 214-unit TR total because the check fires on every criterion unit belonging to a Task-2 essay that is missing its prompt text, not on Task Response units alone (Section 4.2, test 2); a Task-2 essay contributes a CC, LR, and GRA unit alongside its TR unit, and any one of those is refused on the same missing-instruction condition.

Within the 1,188 units that require a visual input, 1,182 were valid and 6 invalid. The frame spanned 379 unique asset identities and 174 unique visual assets, against 1,853 units sharing a duplicate asset.

The decision the harness returned was HOLD: 366 invalid payloads out of 2,232 selected units is a fail-closed condition, and no paid grading call was started against this frame.

Table 5: Refusal codes on the invalid payloads

Refusal code	Count
MISSING_TASK2_INSTRUCTION	360
VISION_IMAGE_REQUIRED	6
Total invalid	366

4.4 HOLD is the working outcome, not a failure

A lifecycle view of automated-system deployment expects measurement and monitoring to gate a deployment claim, not follow it [18], and the evaluation framework this report follows treats a precision-and-reliability check of this kind as its own stage, separate from any score-quality claim [10]. Read against that framing, HOLD is not an incident needing explanation, and it is not an accuracy result: it means zero criteria were scored on this run. It is the harness doing the job it was built for: naming 366 units by refusal code rather than letting them enter a paid run unflagged, and holding the batch until MISSING_TASK2_INSTRUCTION and VISION_IMAGE_REQUIRED are cleared at the source, not patched around downstream.

5 LLM-as-judge with frozen evidence and typed exits

Some questions in the engine’s evaluation and QA machinery cannot be answered by a deterministic check: they require a judgement call over evidence that is itself contestable. For this class of question the engine uses an isolated judge: a fresh-context language model instance with no memory of the conversation that produced the artefact under review, asked exactly once and given no route back into the conversation.

The mechanism is built to defeat two known failure modes of LLM self-review: evidence moving underneath the judge during the call, and a judge asked repeatedly until it produces the wanted answer. Before the call, every evidence file the judge will read is frozen and hashed with SHA-256. The judge is asked the question exactly once. After the call returns, the same evidence files are re-hashed. If any hash has changed, the judge’s own text is discarded outright, regardless of content, and the verdict is forced to HOLD(STALE_EVIDENCE). This is a mechanical contamination check on the evidence window, not a review of what the judge wrote.

The caller sees this as a typed exit code:

Table 6: Isolated-judge exit codes

Exit code	Meaning
0	OK: answered inside the frozen evidence window, no hash drift
10	HOLD(STALE_EVIDENCE): an evidence file’s hash changed between the pre-call and post-call read
11	TIMEOUT: the process tree was killed after the default 180-second budget
12	ERROR: the judge process itself exited non-zero

Tool access is denied structurally, at process launch, rather than by prompt instruction the model could disregard: the judge is started with an empty tool list, strict MCP configuration

restricting any external server, user-only setting sources, and a fresh temporary working directory it has never seen before. The judge can read only the frozen evidence it is handed.

This follows documented precedent in the LLM-as-judge literature. Zheng et al. report that LLM judges carry measurable position, verbosity and self-enhancement biases, the basis for asking once against frozen evidence rather than letting a judge iteratively re-grade its own or another model’s output [12]. Liu et al.’s G-Eval shows that constraining a judge to structured, form-filling output improves alignment with human judgement over free-form critique, the template this mechanism follows for what the judge is permitted to return [13].

The isolated judge adjudicates process and evidence integrity only, for example whether a set of counts is internally consistent or an artefact is what it claims to be. It never scores an essay and never emits a band, a MET/UNMET verdict, or any other grade label; that boundary is held by the same architecture that keeps the deterministic scorer, not the generative model, as the sole source of a result.

6 Model training and cost (QLoRA on a 27B open-weight model, preflight, throughput and cost per real token)

This section reports how the generative model was fine-tuned and what that training measurably cost. Every number below is a configuration value or a direct measurement, not a capability claim, and none of it is an accuracy or MAE result for the engine.

6.1 Base model and precision profile

The fine-tuned base is a 27-billion-parameter open-weight model, referenced in the project’s own training configuration as `Qwen/Qwen3.8-27B` under recipe id `qwen38-27b-cp2-nf4-20260822`. This identifier is drawn from the project’s local training configuration and the downloaded model’s own local model card; it could not be independently confirmed against a public model registry within this session (the web-search budget available to this report was exhausted on other checks), so it is reported here as the project’s internal record of the base model, not as an independently verified public release, and Section 9 flags it for a human check alongside the ISO/IEC citation below. The active precision profile is NF4 QLoRA: base weights are quantised to 4-bit NormalFloat (NF4) while the LoRA adapter arithmetic runs in BF16. This follows the QLoRA method [19], which fine-tunes a quantised base model through low-rank adapters [20], implemented here with the Hugging Face PEFT library [21].

6.2 LoRA configuration

Table 7: LoRA configuration

Parameter	Value
Rank	16
Alpha	16
Target modules	all-linear, text decoder only
Vision tower and aligner	frozen
Loss	completion-only, chunked Triton weighted implementation

Only the text decoder is adapted. The vision tower and its aligner stay frozen throughout, so the vision path that Task Achievement grading depends on is not itself being retrained by this recipe.

6.3 Preflight gate

Before any paid QLoRA step is allowed to run on rented hardware, a lockfile stage checks that every optimisation lever declared for the run is present and wired into the trainer, then signs a hashed per-launch manifest recording each lever’s value and its source location. Training spend on a rented GPU is not authorised until this gate passes.

6.4 Measured throughput and cost

Table 8: Measured training throughput and cost per rental GPU rail

Rail	Throughput	Cost per million real tokens
RTX 5090	approximately 944 nonpad tokens/sec at physical batch 1	\$0.12
RTX PRO 6000	not separately reported	\$0.21
B200	not separately reported	\$0.55

Batch size on the RTX 5090 rail is fixed at 1. The observed, reproduced failure at batch 2 is a vision-cache identity-verification defect (a cache-versus-live mismatch error), not an out-of-memory event. Separately, and not itself an observed failure, the rail’s memory headroom at batch 1 (about 25GB used of 31.4GB available) leaves little margin to double activations, which is why batch 1 is treated as the only rail configuration proven safe to run, even though the specific batch-2 failure actually seen was the cache defect rather than an out-of-memory error. These are single-run, single-host measurements: no repeated-run variance is reported, and no non-rental cloud-API baseline was measured alongside the three rental rails, so the cost figures below should be read as one project’s infrastructure economics on this specific setup, not as a general claim about training cost. On a per-real-token basis, the RTX 5090 rail is stated as 4.5 times cheaper than the B200 rail in this measurement.

6.5 Inference-serving throughput

Separately from the training measurement above, an inference-serving throughput check on vLLM found that enabling CUDA graphs, rather than eager execution, raised measured throughput from 108 to 261 tokens per second, a 2.4 times gain, with vision-plus-LoRA graph capture stable. This is a serving-layer result, not a training result, and like the figures above it carries no accuracy or MAE claim.

None of the numbers in this section, throughput, cost per real token, or the CUDA-graph serving gain, measure how well the model grades an essay. They describe training configuration and infrastructure economics only; the current four-criterion production-engine MAE remains MISSING, as stated elsewhere in this report.

7 Operations: batch ladder, receipts, data-quality gate, drift detector, guardrail classifier scored on pinned decks

7.1 Batch ladder

Production and confirmation batches climb a fixed three-rung ladder before reaching full width: $n = 1$, then $n = 5$, then $n = 30$, then full width. Each rung must be proven clean before the next is allowed to fire. Confirmation lanes (re-checking rows against an already-settled bar) and generation lanes (minting a new label) are distinct instruments for this purpose: each requires its own $n = 1$ certification, and a pass on one does not stand in for a pass on the other.

7.2 Receipts

Every write to a shared data partition or overlay is gated by a two-state receipt. A PREPARED receipt is issued only against a non-empty authorisation token and records `rows_before`, the row count immediately ahead of the write. A COMMITTED receipt is issued only after `rows_after` (the row count following the write) and `rows_delta` (`rows_after` minus `rows_before`) have been computed and verified against the actual bytes written to disk, not against an expected value. The human-readable summary line for a commit names all of these together in one place: `authorized_by, rows_before, rows_after, rows_delta`, and a backup hash.

7.3 Data-quality gate

The automated data-quality gate runs 44 checks against the corpus. Its most recently recorded run:

Table 9: Most recent data-quality gate run

Metric	Value
Total checks	44
Pass	25
Fail	19
Critical	7
Auto-fixable	1
Needs review	9
Info	2
Exit code	2

The gate’s documented exit-code contract is: 0, every check passed; 1, one or more needs-review items are pending a human decision; 2, at least one critical issue was found. The seven checks recorded at critical severity, and their current flagged status, are: cross-source normalised duplicate groups (1,533 rows, flagged for quarantine); a small word-count and Task-Response-score anomaly (3 rows, flagged for quarantine); null Task Response trait scores from one named source, recorded as two related checks (64 rows each); cross-split evaluation contamination (1,554 row pairs, 1,546 of them across a train/test or train/staging split boundary, flagged for quarantine); duplicate essay identifiers (13 duplicate-ID groups); and foreign-language contamination in essay text (86 rows, including 5 rows on the human-rater reference set). None of these seven is reported here as fixed; they are named so this report does not cite an admittedly-failing gate as a clean pass, and their resolution is tracked in the underlying gate report, not reproduced further in this document.

For scale, the corpus the gate runs against:

Table 10: Corpus scale

Dimension	Split
Rows by columns	99,455 rows, 105 columns
Corpus size, live	101,175 essays across 47 distinct sources, as of this writing
Provenance	human 75,018 / synthetic 24,437
Domain	<code>ielts_t2</code> , <code>ielts_t1</code> , <code>off_domain</code> , <code>ielts_gt_letter</code>
Examiner comments, mined to distilled	2,014 mined → 264 label-free per-criterion training facts (no TA tier ex
Cold-start SFT bundle	9,825 rows, label-free

The 99,455-row snapshot above is the QA gate’s own dated corpus count; it predates the live corpus figure. As of this writing the corpus stands at 101,175 essays across 47 distinct sources, 1,688 more than this report’s most recent written snapshot of 99,487, per the live measurement tool rather than either snapshot.

A separate weak-supervision pool carries raw band labels and is deliberately excluded from the default training path, to keep raw grades out of the model. Its size is 223,600 rows: confirmed by direct line count of the file itself and independently reproduced by re-running the assembly script’s own row-selection filter against the source parquet (57,078 filtered essays, up to five per-criterion labels each, summing exactly to 223,600). The competing 99,000/“99k” figure in the assembly script’s own header comment is a stale approximation that does not match either the file or its own filter’s essay count, and is corrected here rather than left as an open conflict. The on-doctrine substitute actually used for training is 4,248 label-free contrastive pairs, which is unaffected by this correction.

7.4 Drift detector

A record-vs-reality drift detector is described here at the design level: it checks whether a document that declares itself the current source of truth is still current against live state, and whether a gate still enforces a rule the governing record has since retired. No run-count or hit-rate figure for the drift detector itself is present in the evidence record, and none is claimed here; this follows the project’s anti-fabrication rule, under which missing data is written as missing rather than estimated. The function sits under the monitor and manage stages of a standard AI risk-management lifecycle [18].

7.5 Guardrail classifier

A frozen, 1,219-case policy and spend deck scores the guardrail that sits ahead of write and spend actions. Replayed against the baseline gate, the deck produced 174 false positives (198 weighted) and 28 false negatives. Replayed against the identical deck, a replacement classifier produced 51 false positives and 0 false negatives, meeting the pre-registered bar of fewer than 174 false positives and exactly zero false negatives:

The replacement classifier also carries its own 30-case must-deny selftest deck (10 spend, 10 irreversible, 10 label-leak cases) plus 5 clean-allow cases, with zero recorded selftest failures.

Table 11: Guardrail classifier: baseline vs. replacement, replayed against the identical 1,219-case deck

Classifier	False positives	False negatives
Baseline gate	174 (198 weighted)	28
Replacement gate	51	0

7.6 Posture

These mechanisms are presented under a documented AI management-system posture; that posture is flagged in Section 9 for independent confirmation [22]. The governing evaluation framework treats operational monitoring, not a one-time score, as the evidence that a validity claim continues to hold in production, which is why this section reports receipts and gate runs rather than a single accuracy figure [10].

8 Measured development-run results

Separately from the current architecture’s own unmeasured engine MAE (Section 9), the project’s development record holds measured per-criterion MAE, QWK, and rho figures from two earlier, now-superseded scoring-mechanism generations: a theta-nearest-neighbor scorer with an LLM-ranking Task 1 head (V11.3), and a separate Task 2 LLM-retrieval scorer (T2-DTAOI), both from June 2026. These are real, dated measurements, not measurements of the label-free, deterministic-Rasch pipeline this report documents elsewhere: neither source system describes an LD1 label quarantine, a deterministic Rasch scorer, or a vision-law-compliant image requirement. They are reported here in full, scope and session named beside every figure, per this project’s own rule that every number states what it is out of; where the record itself conflicts, both values are printed rather than one being silently adopted.

Table 12: Measured development-run results, by system generation (historical, superseded)

System generation	Split and n	Criterion	MAE	QWK or ρ	Session	Note
V11.3 (theta-kNN scorer, T1 DTAOI vision-ranked)	gate_spot30, $n = 30$, essays with a precomputed ability coordinate	GRA	0.083	QWK 0.9946	s132-134 (Jun 2026)	Superseded architecture; theta coverage was approximately 53.5% of the corpus at the time
V11.3	same	LR	0.100	QWK 0.9825	s132-134	Superseded architecture
V11.3	same	CC	0.083	QWK 0.9894	s132-134	Superseded architecture
V11.3	same	TA	CONFLICT: 0.2333 vs 0.333	QWK 0.9746 (0.2333 source)	s132-134	Two dated internal sources disagree; both printed, not resolved

Table 12: Measured development-run results, by system generation (historical, superseded)

System generation	Split and n	Criterion	MAE	QWK or ρ	Session	Note
V11.3, theta-only ablation (no vision ranking)	same $n = 30$ pool, k-NN on theta alone	TA	0.417	–	s132-134	Ablation removing the DTAOI vision-ranking component
V11.3, blended/overall	gate_spot30 summary figure	Blended (all criteria)	0.15	–	s132-134	Different metric shape than the per-criterion rows above
V11.3, essays without a precomputed coordinate	fallback path, majority of new/unseen essays at the time	All	0.6-0.8	$\rho \approx 0.35$	s132-134	Fallback coverage: roughly the other half of the corpus, outside the $n = 30$ gated holdout above
T2-DTAOI production gate	$n = 157$, vault-calibrated production gate	TR	0.319	ρ 0.941	s136	PASS
T2-DTAOI production gate	$n = 157$	CC	0.268	ρ 0.915	s136	PASS
T2-DTAOI production gate	$n = 157$	LR	0.248	ρ 0.919	s136	PASS
T2-DTAOI production gate	$n = 157$	GRA	0.226	ρ 0.921	s136	PASS
T2-DTAOI cross-source NSC label refinement	TR only, before vs after refinement	TR	1.29 to 0.95 (26% better)	–	s136	Labelled “cross-source” in its own source
T2-DTAOI per-band vault isotonic calibration	$n = 65$ vault essays	TR	0.308 vs 0.446 global-offset baseline (-31%)	–	s137	Band 4.5 slice 0.500 vs 0.786 (-36%); band 6.5 slice 0.500 vs 1.125 (-56%)

Both source systems are architecturally distinct from, and predate, the label-free extract-then-Rasch-score pipeline this report documents (Sections 2 to 7); neither result is a measurement of the engine described there, and neither substitutes for the current architecture’s MISSING four-criterion MAE (Section 9).

9 Limitations and what is deliberately not claimed

The central limitation is stated first because everything else in this section follows from it: the current four-criterion production-engine MAE (TA, CC, LR, GRA, matching the scope of the

measured human-rater floor in Section 3) is MISSING. TR, the fifth criterion, has no human-rater floor in the record and is separately unmeasured. No accuracy, MAE, agreement-with-examiner, or band-prediction performance claim is made for the engine, on any subset of criteria, anywhere in this report. The one figure sometimes quoted for this project, an MAE of 0.61, was measured on GRA alone and is retired: it is LEGACY-UNVERIFIED and is cited nowhere in this report as a current result. The only measurement this report treats as current is the fail-closed harness’s dry-run outcome, a HOLD decision over a 2,232-unit frame, which stopped the pipeline before any paid measurement run began and which measured the harness’s own controls, not engine accuracy.

This report is a systems and evidence report, not an accuracy evaluation paper, and it is structured as one: Sections 2 to 4 report what the architecture guarantees by construction, Sections 5 to 7 report measured operational and infrastructure evidence (the isolated-judge mechanism, batch and receipt controls, the data-quality gate, and the guardrail classifier deck), and no section substitutes for the missing engine-accuracy measurement named above.

No baseline comparison has been run. Section 10 names architectural departures from feature-based AES, neural end-to-end AES, and the ASAP-AES/QWK benchmark tradition, but no naive baseline (for example a majority-band or length-based regression) and no untuned, off-the-shelf LLM grader has been scored on this project’s own held-out set under the same per-criterion MAE metric. The departures named in Section 10 are architectural and design differences, not measured performance comparisons.

The abstain mechanism’s two thresholds (Section 2.6) are configured design values. No abstain rate and no coverage-versus-accuracy curve has been measured against them at production scale, because that measurement depends on the same engine-level run that produced the MISSING MAE above.

The grade-label firewall’s scope is narrower than a blanket guarantee. It is a disk-artefact scanner covering specific generation-lane file classes, applying value-carrying regex patterns only, never a bare prohibition word. It has not been tested against a paraphrased or semantic leak that avoids its literal matched tokens. A clean scan is not proof of absence across every possible leakage surface, and the scanner excludes the student-facing release surface by design; this report does not audit that surface.

The harness carries an internal documentation discrepancy that this report discloses rather than resolves. Its own runbook records a test tally of six tests passed, but the test file it describes now defines eight test functions. The report leaves this open rather than silently picking the newer count as authoritative, because neither figure has been re-verified live in this session.

The weak-supervision pool size, previously reported inconsistently as 99,000 rows against 223,600 rows (Section 7.3), is confirmed at 223,600 rows by direct file count and independent recomputation from source; the 99,000 figure was a stale code-comment approximation, now corrected.

The base model identifier reported in Section 6.1 is drawn from this project’s own training configuration and local model card and could not be independently confirmed against a public registry within this session’s remaining search budget; it is flagged for a human check before being relied on as a public-release claim.

The exact token, cost and runtime figures for the $n = 1$, $n = 5$ and $n = 30$ rungs of the eventual paid measurement run are MISSING in the plan’s own text and are not estimated here. The cost and throughput figures reported in Section 6.4 are single-run measurements on rented hardware with no repeated-run variance and no non-rental cloud-API baseline.

ISO/IEC 42001 is cited in this report for organisational framing only. Its standard page could not be independently confirmed live in this session, owing to a bot wall on the request, and it is flagged here for a human check before it is relied on as a compliance statement [22].

This report holds itself to the standard that a validity claim for an automated scoring system

requires documented evidence across multiple dimensions, not a single number [10, 14]. Any future per-criterion MAE for this engine will be reported against the measured four-criterion human-rater floor of 0.36, scoped to the same four criteria that floor measures, not against the quadratic-weighted-kappa convention used by the field’s standard leaderboard benchmark [7].

Full evidence sourcing for every claim above, and throughout this report, is retained in an internal evidence map (E1 to E46, resolved to file and line) held privately by the author; it is not included in this document because it consists of private repository paths, and is available for audit on direct request (Appendix A).

10 Related work

This report’s design sits inside four literatures: feature-based AES, neural end-to-end AES, the ASAP-AES/quadratic-weighted-kappa (QWK) benchmark tradition, and LLM-as-judge and efficient-fine-tuning methodology. Table 13 gives each family and its architectural departure; no comparative accuracy number is claimed against any family below, for the reasons stated in Section 9. The paragraphs after the table carry the reasoning, then close on the governing standards and rubric source the design answers to.

Table 13: Related-work families and architectural departures (design differences, not measured performance comparisons)

Family	Representative work	Architectural departure in this report
1. Feature-based AES	e-rater’s grammar/usage/mechanics/style/organisation/development pipeline feeding a regression [2, 3]	Fact-extraction-plus-deterministic/scorer split
2. Neural end-to-end AES	LSTM/CNN and embedding-based scorers [4, 5]; field survey [6]	Generative model never regresses a score
3. Benchmark tradition	ASAP-AES/QWK comparisons [7–9]; kappa statistics [15–17]	Per-criterion MAE against a measured human floor (not yet measured for the engine, Section 9)
4. LLM-as-judge and fine-tuning	Judge/eval methodology [12, 13]; QLoRA lineage [19–21]	Load-bearing for Sections 2, 5 and 6, not contrasted

Family one. e-rater is the reference architecture for feature-based AES: a fixed pipeline extracts grammar, usage, mechanics, style, organisation and development features, and a regression maps that vector to a score [2, 3]. Gnomon keeps the two-stage shape, extraction then scoring, but changes what each stage does. The generative model extracts and quotes facts rather than computing engineered features, across all five IELTS criteria: Task Achievement, Task Response, Coherence and Cohesion, Lexical Resource, and Grammatical Range and Accuracy. A separate deterministic scorer converts those facts into a result: it treats MET/UNMET evidence items as inputs to a Rasch (item-response-theory) model, difficulty estimated label-free from the response matrix, with floor items acting as non-compensatory caps on the band. No regression over surface features runs anywhere in the pipeline. This is a structural departure, not a measured one: no comparative MAE or QWK figure against e-rater or any other feature-based system has been run.

Family two. Neural end-to-end AES trains a network, LSTM- or CNN-based, or built on

score-specific embeddings, to map an essay directly to a holistic score [4, 5]; Ke and Ng’s survey situates both lineages inside AES research [6]. Gnomon’s departure is structural: the generative model is never trained or permitted to regress a score end to end. Its output stays confined to facts and reasoning; the Rasch scorer above is the only component producing a band, per criterion.

Family three. The ASAP-AES competition and studies built on it established QWK as the field’s de facto statistic [7–9], itself a chance-corrected, ordinal-weighted descendant of Cohen’s kappa and intraclass correlation [15–17]. This report does not adopt QWK; its metric arc is per-criterion mean absolute error (MAE) across the five criteria against a measured human-rater floor, not a single holistic agreement figure. That floor currently covers four of the five criteria (Section 3), and no engine-level MAE, on any criterion count, has been measured against it (Section 9).

Family four. LLM-as-judge and structured-generation methodology, MT-Bench’s evaluator-agreement protocol and G-Eval’s chain-of-thought form-filling, informs how this report structures a model’s fact-and-reasoning output [12, 13]. The QLoRA/LoRA/PEFT lineage informs how the generative model is fine-tuned [19–21]. Both are load-bearing rather than contrastive: methodology Sections 2, 5 and 6 build on, not lineages this report departs from.

The design answers to governing standards: Williamson, Xi and Breyer’s AES evaluation framework, the AERA/APA/NCME testing standards, the NIST AI Risk Management Framework, and ISO/IEC 42001 [10, 14, 18, 22]; and to the IELTS band descriptors and companion assessment-criteria guide as the primary rubric source [1, 11]. Across all four families, no prior published AES system known to this report quarantines grade labels from the generative model as a structural property, preventing it from ever emitting a band, MET/UNMET verdict, or score. That quarantine is this project’s principal architectural departure from the surveyed lineages; it is a design claim, not a performance claim, and Section 9 states plainly that the performance question remains open.

All twenty-two references cited in this report were live-checked in the authoring session (2026-09-07); twenty-one resolved with a confirmed HTTP 200 response and one, ISO/IEC 42001:2023, did not (HTTP 403, Cloudflare bot wall) and is flagged in the bibliography entry for a human re-check rather than reported as verified.

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Appendix A: Evidence and Data Availability

Every numbered claim in this report (evidence identifiers E1 through E53) resolves to a specific file-and-line location in the project’s private repository. That mapping is retained internally as an evidence map and is not reproduced in this document, because it consists of private repository paths, internal session identifiers, and internal project codenames that this report’s own drafting rules exclude from external distribution. The evidence map is available for direct audit on request.

Items marked MISSING (verbatim, nothing substituted)

- Current four-criterion production-engine MAE: MISSING, pending vision-compliant re-measurement (Section 9).
- Token, cost and runtime figures for the $n = 1$, $n = 5$ and $n = 30$ measurement rungs: MISSING.
- Weak-supervision pool row count: RESOLVED (previously reported inconsistently as 99,000 vs 223,600, Section 7.3) – confirmed 223,600 rows by live file count and independent recomputation from the source parquet filter; 99,000 was a stale code-comment approximation, not a genuine second count.
- The current resolution status of the seven critical data-quality checks named in Section 7.3, beyond the flagged counts already reported: not reproduced further in this document; tracked in the underlying gate report.